

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Hard

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

Compound **X** is produced from cyclohexene. An amine which is a relatively weak base containing one nitrogen atom can be produced from compound **X** in one step.

Outline a mechanism for the formation of compound **X** from cyclohexene. You will need to give suitable reagents in your answer.

[3 marks]

Question 1b

One condition that is needed to convert compound **X** into the amine molecule is that the reagent needs to be in excess.

ii)

State the reagent and any other required conditions for the reaction.

[2]

ii)

Suggest why the reagent needs to be in excess.

[1]

[2 marks]

Question 1c

Propylamine acts as a Brønsted-Lowry base when it reacts with water.

Suggest an equation to show how propylamine reacts when mixed with water.

[1 mark]

Question 2a

2-hydroxy-2-phenylpropanenitrile is shown in Fig. 2.1.

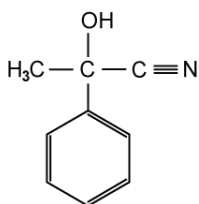


Fig. 2.1

Deduce the number of sigma (σ) and pi (π) bonds in 2-hydroxy-2-phenylpropanenitrile.

sigma

pi

[2 marks]

Question 2b

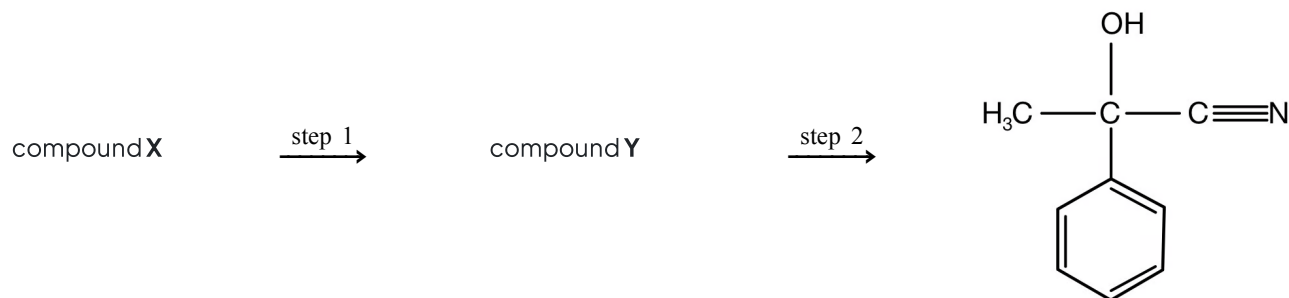
A molecule of 2-hydroxy-2-phenylpropanenitrile contains sp , sp^2 and sp^3 hybridised carbon atoms.

State the number of sp , sp^2 and sp^3 hybridised carbon atoms in a molecule of 2-hydroxy-2-phenylpropanenitrile.

[3 marks]

Question 2c

2-hydroxy-2-phenylpropanenitrile can be produced from compound X, a secondary alcohol, in a two-step synthesis.



State the reagents and conditions required for step 1 and step 2.

reagents for step 1

conditions for step 1

reagents for step 2

conditions for step 2

[4 marks]

Question 2d

i)
Draw the mechanism for the second step in the synthesis of 2-hydroxy-2-phenylpropanenitrile.

You should clearly show the structure of compound **Y** and any intermediate formed and include all relevant charges, partial charges, curly arrows and lone pairs.

[6]

ii)
Draw the structure of compound **X**.

[1]

[7 marks]**Question 3a**

2,2-dimethylpentanenitrile is useful in the synthesis of a variety of medicines and pharmaceuticals.

Draw the skeletal formula of 2,2-dimethylpentanenitrile.

[1 mark]

Question 3b

2,2-dimethylpentanenitrile undergoes hydrolysis when heated with dilute hydrochloric acid.

Write an equation for the hydrolysis of 2,2-dimethylpentanenitrile.

[2 marks]

Question 3c

2,2-dimethylpentanoic acid can also be produced by the hydrolysis of 2,2-dimethylpentanenitrile using sodium hydroxide but this occurs in two steps.

i)

Draw the fully displayed formula of the intermediate organic product.

[1]

ii)

State the other product formed with this intermediate organic product.

[1]

iii)

State the type of reaction that is required to produce 2,2-dimethylpentanoic acid from this intermediate organic product.

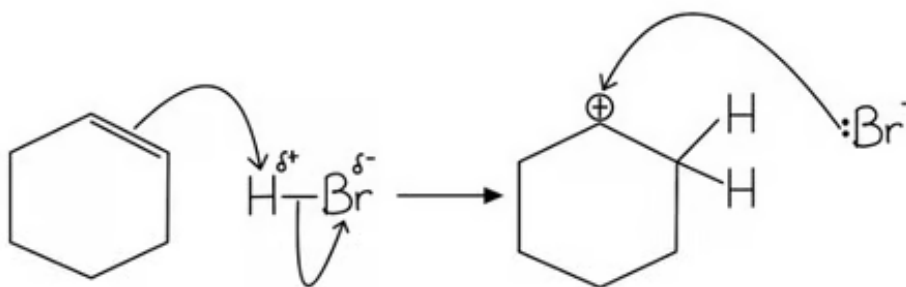
[1]

[3 marks]

Model Answers: Hard

1a

a) The mechanism for the formation of compound **X** is:



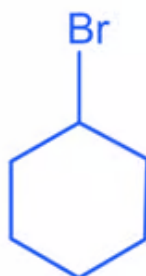
- Curly arrow from C=C to δ^+ H atom; [1 mark]
- Curly arrow from H-Br bond to δ^- Br atom; [1 mark]
- Correct carbocation drawn

AND

Curly arrow from :Br^- to positively charged carbon atom; [1 mark]

[Total: 3 marks]

- You could have used H-Br, H-Cl or H-I as the reagent to gain these marks
- If using HBr, the structure of compound **X** is:



- Remember that one way to form an amine molecule is for a halogenoalkane to undergo nucleophilic substitution
- Formation of a halogenoalkane from an alkene would involve electrophilic addition

- The product contains only one nitrogen atom, so the reagent must be a hydrogen halide molecule, and not a halogen molecule

1b

b)

i) The reagents and conditions for the conversion of compound **X** into the amine molecule are:

- Reagent: Ammonia / NH_3 ; [1 mark]
- Conditions: Ethanolic / ethanol as a solvent

AND

Under pressure / in a sealed tube; [1 mark]

ii) The reagent needs to be in excess to:

- Prevent amine attacking compound **X** / halogenoalkane

OR

Prevent the formation of secondary amines; [1 mark]

[Total: 3 marks]

- Nucleophilic substitution is the mechanism for this conversion and in order to make sure the favoured amine is produced, in this case phenylamine, excess ammonia is required
- This also must take place under pressure or in a sealed container so do not forget to give the conditions when asked
- If ammonia is not in excess, the amine can act as a nucleophile and attack the halogenoalkane, compound **X**
- This would cause the formation of secondary amines in which the nitrogen atom is attached to 2 alky/aryl groups
- You do not need to know about secondary or tertiary amines for AS level, but you should recognise that the nitrogen on the amine has a lone pair of electrons that can act as a nucleophile
- Combined with the knowledge that halogenoalkanes undergo nucleophilic substitution should lead to the suggestion that the amine is able to attack compound **X**

1c

c) An equation to show how propylamine reacts with water is:



[Total: 1 mark]

- A Brønsted-Lowry base is a species that accepts a proton (a hydrogen ion)
- In this case, propylamine will accept a hydrogen ion from water
- Water acts as a Brønsted-Lowry acid as it donates a hydrogen ion
- Amines are weak bases which can be shown using the reversible sign, however, you would gain the mark if you used an arrow instead of the reversible sign

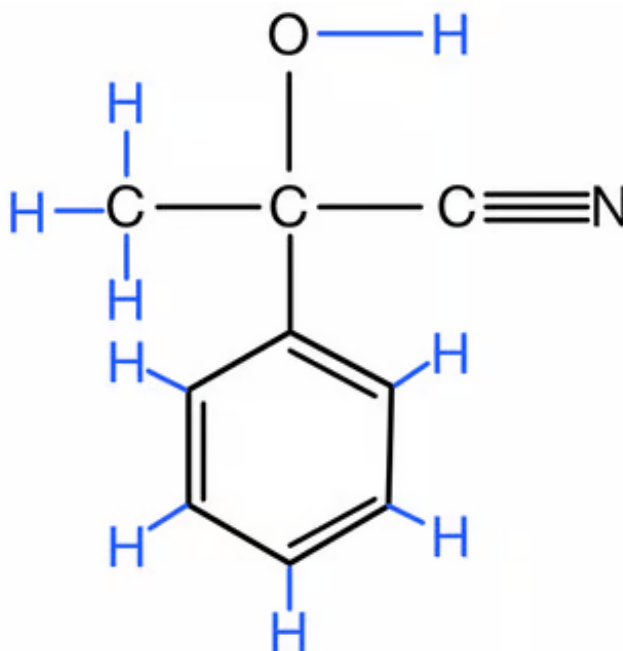
2a

a) The number of sigma (σ) bonds and pi (π) bonds in 2-hydroxy-2-phenylpropanenitrile is:

- Sigma = 20; [1 mark]
- Pi = 5; [1 mark]

[Total: 2 marks]

- Careful:** It is very easy to miss sigma bonds in the structure because:
 - Some aren't shown as they have been written into the CH_3 and OH groups
 - There are 5 "hidden" hydrogens in the phenyl group of the structure
- Tip:** Re-draw the molecule in a way that includes the "hidden" hydrogen atoms AND highlights the bonds that are within CH_3 and OH groups

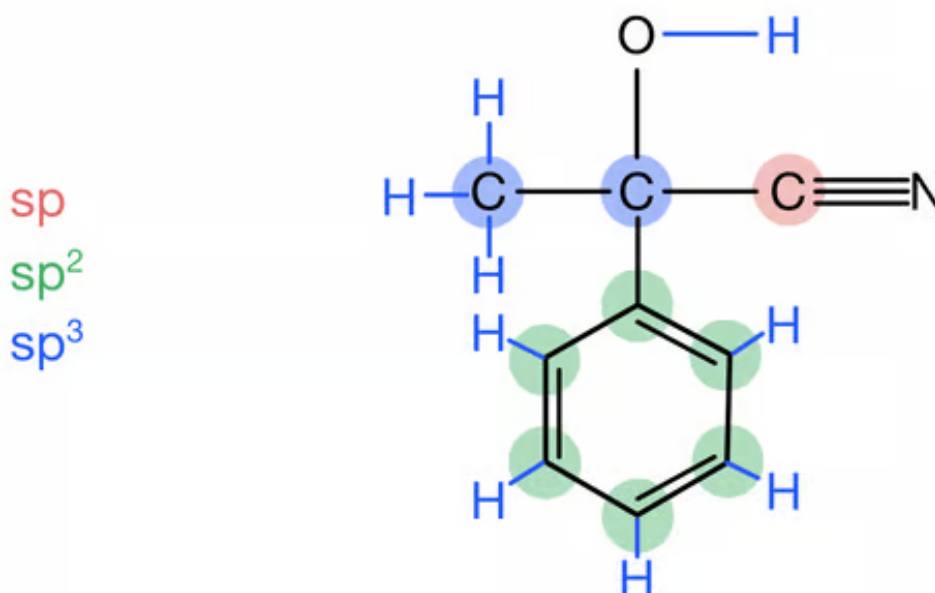


- **Remember:**
 - Single bonds are sigma bonds
 - Double bonds are one sigma and one pi bond
 - Triple bonds are one sigma and two pi bonds
- There are 20 bonds in total
 - Therefore, there are 33 sigma bonds
- There are 3 double bonds, which have a pi bond each, and 1 triple bond, which has 2 pi bonds
 - Therefore, there are 5 pi bonds

2b

b) The number of sp , sp^2 and sp^3 hybridised carbon atoms in a molecule of 2-hydroxy-2-phenylpropanenitrile is:

- $sp = 1$; [1 mark]
- $sp^2 = 6$; [1 mark]
- $sp^3 = 2$; [1 mark]



2c

c) The reagents and conditions required for step 1 and step 2 are:

- Reagents for step 1: Acidified potassium dichromate **OR** $K_2Cr_2O_7$ **AND** H^+ / H_2SO_4
OR
Acidified potassium manganate **OR** $KMnO_4$ **AND** H^+ / H_2SO_4 ; [1 mark]
- Conditions for step 1: Heat; [1 mark]
- Reagents for step 2: HCN and KCN (as a catalyst); [1 mark]
- Conditions for step 2: Heat; [1 mark]

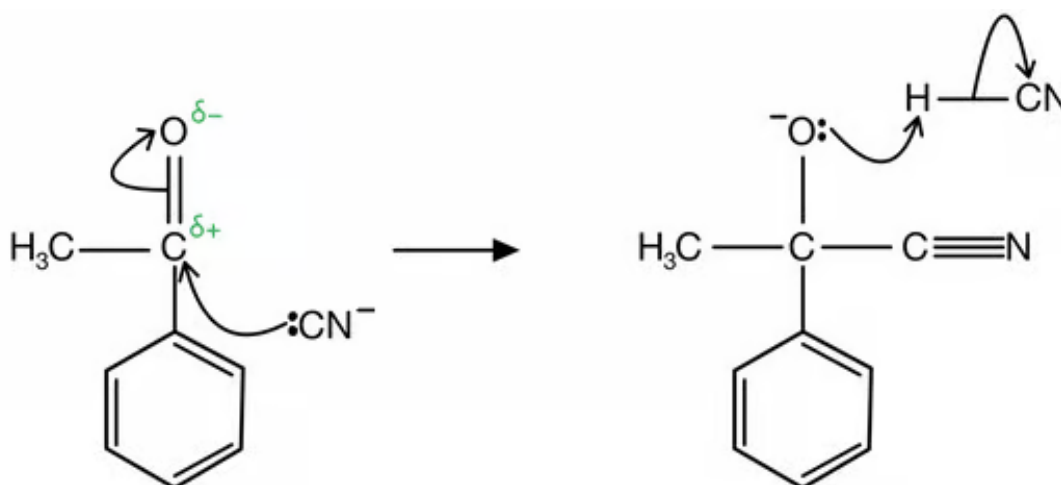
[Total: 4 marks]

- This type of question shows how important it is to learn the different routes of organic synthesis and the reagents and conditions necessary for each route
- A hydroxynitrile can be synthesised from an aldehyde or ketone by nucleophilic addition with HCN
- HCN is produced in situ by the reaction of KCN and dilute sulfuric acid
- 2-hydroxy-2-phenylpropanenitrile would be produced from a ketone
- Ketones are produced by heating a secondary alcohol with an oxidising agent
 - Distillation is not required as further oxidation of the ketone cannot occur

2d

d)

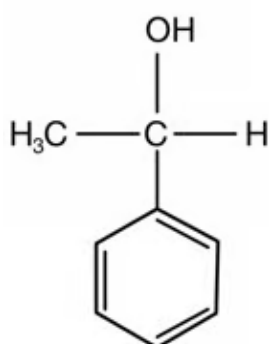
i) The mechanism for the second step in the synthesis of 2-hydroxy-2-phenylpropanenitrile is:



- Correct structure of compound **Y**; [1 mark]
- Curly arrow from lone pair on C of :CN⁻ to δ⁺ C atom; [1 mark]
- Curly arrow from C=O bond to δ⁻ O atom; [1 mark]
- Correct structure of intermediate including a negative charge on O atom; [1 mark]
- Curly arrow from lone pair on O atom to H of HCN; [1 mark]
- Curly arrow from H-CN bond to C atom; [1 mark]

ii) The structure of compound **X** is:

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; [1 mark]

[Total: 7 marks]

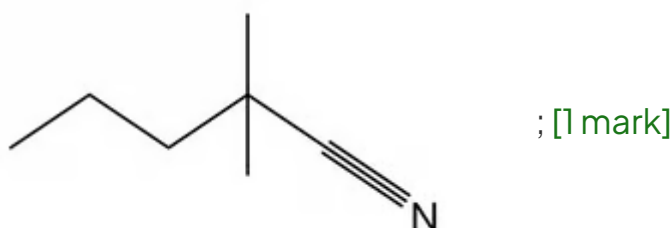
- **Remember:** The carbon chain lengthens when this hydroxynitrile is produced, so compound **Y** will have one less carbon
- Make sure you show all lone pairs and charges (both full and partial)

- You do not need to show the end product in the mechanism
- The CN^- ions come from KCN, these are regenerated, so KCN acts as a catalyst
- Compound **X** is secondary alcohol which is oxidised to produce compound **Y** so its structure can be drawn by replacing $\text{C}=\text{O}$ with a $\text{C}-\text{O}$, adding an H to the O and adding a $\text{C}-\text{H}$

3a

a) The skeletal formula of 2,2-dimethylpentanenitrile is:

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[Total: 1 mark]

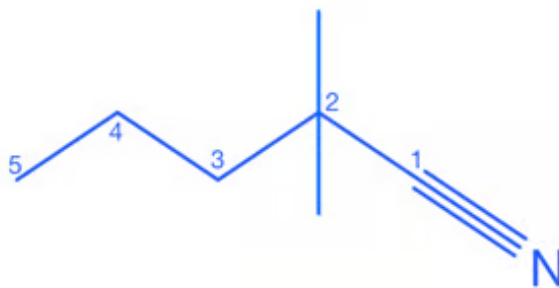
- To draw the skeletal formula of 2,2-dimethylpentanenitrile:
 - Draw the straight chain as indicated by the stem of the name -pentane



- Add the nitrile functional group
 - As this is a $\text{C}\equiv\text{N}$, this must be on an end carbon, this carbon is therefore carbon 1



- Add the two methyl groups on carbon 2:



- **Careful:** It is a common mistake for students to add the nitrile group in addition to the carbons present in the main carbon chain but you must remember that the carbon in the nitrile group is part of the carbon chain

3b

b) An equation for the hydrolysis of 2,2-dimethylpentanenitrile is:



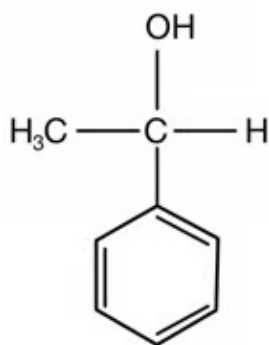
- Correct products; [1 mark]
- Correct reactants

AND

balanced; [1 mark]

[Total: 2 marks]

- You would obtain the marks if you had drawn the displayed formula of 2,2-dimethylpentanenitrile and the organic product
- Hydrolysis is the breakdown of a compound using water, so you must include water as a reactant
- Hydrolysis by dilute acid results in the formation of a carboxylic acid and ammonium salt
 - In this case, 2,2-dimethylpentanoic acid and ammonium chloride are produced



; [1 mark]

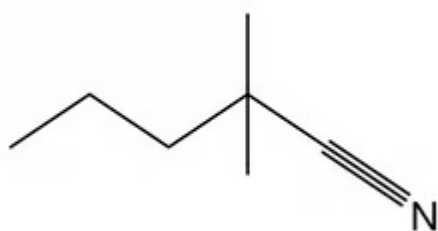
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- **Remember:** The carbon chain lengthens when this hydroxynitrile is produced, so compound **Y** will have one less carbon
- Make sure you show all lone pairs and charges (both full and partial)
- You do not need to show the end product in the mechanism
- The CN⁻ ions come from KCN, these are regenerated, so KCN acts as a catalyst
- Compound **X** is secondary alcohol which is oxidised to produce compound **Y** so its structure can be drawn by replacing C=O with a C-O, adding an H to the O and adding a C-H

3a

a) The skeletal formula of 2,2-dimethylpentanenitrile is:

•



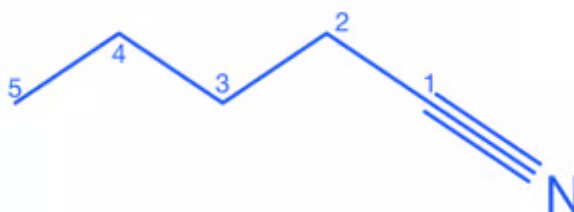
; [1 mark]

[Total: 1 mark]

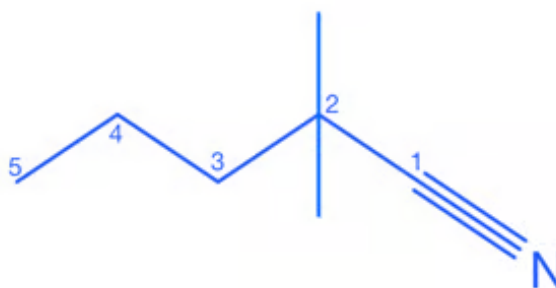
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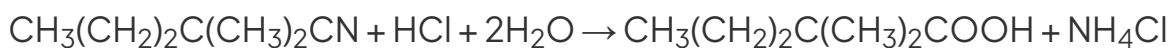
- Add the two methyl groups on carbon 2:



- **Careful:** It is a common mistake for students to add the nitrile group in addition to the carbons present in the main carbon chain but you must remember that the carbon in the nitrile group is part of the carbon chain

3b

b) An equation for the hydrolysis of 2,2-dimethylpentanenitrile is:



- Correct products; [1 mark]
- Correct reactants

AND

balanced; [1 mark]

[Total: 2 marks]

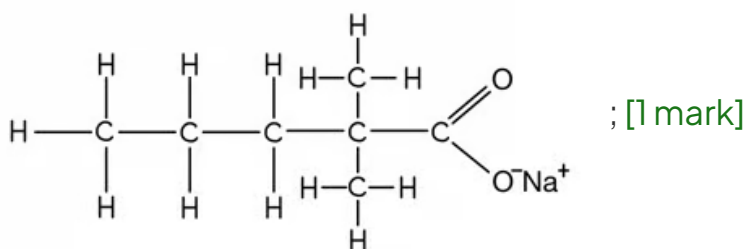
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- Hydrolysis is the breakdown of a compound using water, so you must include water as a reactant
- Hydrolysis by dilute acid results in the formation of a carboxylic acid and ammonium salt
 - In this case, 2,2-dimethylpentanoic acid and ammonium chloride are produced

3c

c)

i) The fully displayed formula of the intermediate organic product is:

•



ii) The other product formed with this intermediate organic product is:

- Ammonia; [1 mark]

iii) The type of reaction that is required to produce 2,2-dimethylpentanoic acid from this intermediate organic product is:

- Acidification; [1 mark]

[Total: 3 marks]

- Sodium hydroxide is an alkali, so alkali hydrolysis will occur